Angle facts and polygons

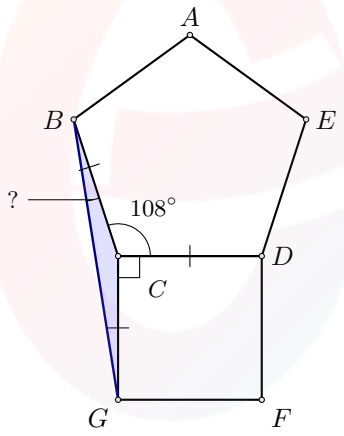
L1-GE1 Level 1 Foundations · Geometry

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How to find angles using lines, parallel lines, triangle and polygon angle sums, isosceles triangles and regular polygons.
By the end you can use angles on lines and at points, parallel lines and triangle angle sums, with reasons; find equal sides and use isosceles triangles to calculate unknown angles; work out interior and exterior angles of polygons and decide which regular polygons fit round a point.
Prerequisites: none. **Contests:** JMC, Junior Kangaroo, JMO, AMC 8. **Time:** about 120 minutes.

1 Lesson

The diagram shows a regular pentagon $ABCDE$ and a square $CDFG$ drawn on its side CD, outside the pentagon. Find $\angle CBG$.



No angle of triangle BCG is given, so we look for facts that produce one. Each angle of a regular pentagon is 108° (proved below) and each angle of a square is 90°. The angles round C add to 360°, so $\angle BCG = 360^\circ - 108^\circ - 90^\circ = 162^\circ$. Next we look at lengths: CB is a side of the pentagon and CG is a side of the square, and both equal CD. So triangle BCG is isosceles with $CB = CG$. Its base angles are equal and add to $180^\circ - 162^\circ = 18^\circ$, hence $\angle CBG = 9^\circ$.

Most angle questions on the JMC, the Junior Kangaroo and the AMC 8 combine a few facts like these three. This unit proves each one and shows how to choose the next one to use.

1.1 Angles at a point and on a line

A full turn is 360°, so the angles round a point add to 360°, and angles on one side of a straight line add to 180°. When two lines cross, the opposite angles are equal; they are called *vertically opposite* angles. Indeed, if the four angles in order are a, b, c, d, then $a + b = 180^\circ$ and $b + c = 180^\circ$, since each pair lies on a straight line, so $a = c$.

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1.2 Parallel lines

Fact 1 (parallel lines).
When a line crosses two parallel lines, corresponding angles are equal (the F shape), alternate angles are equal (the Z shape), and co-interior angles add to 180° (the C shape).

Parallel lines point in the same direction, so the crossing line makes the same angles with each of them; angles in matching positions at the two crossings are the corresponding angles. An alternate angle is vertically opposite a corresponding angle, and a co-interior angle is the neighbour of a corresponding angle on a straight line, which gives the other two facts. Each fact also works in reverse: if alternate angles are equal, then the lines are parallel.

1.3 Triangles

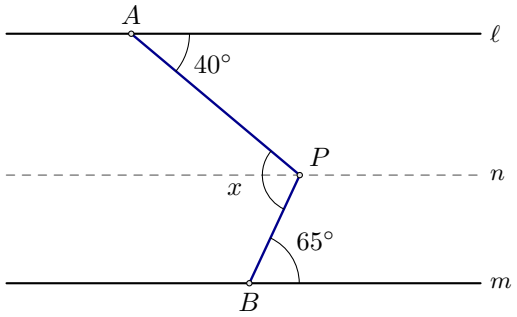
Fact 2 (triangles).
The angles of a triangle add to 180°. An exterior angle of a triangle equals the sum of the interior angles at the other two vertices.

Proof. Take a triangle ABC and draw the line through A parallel to BC. The angles this line makes with AB and AC are alternate to $\angle B$ and $\angle C$, and together with $\angle A$ they fill a straight line, so $\angle A + \angle B + \angle C = 180^\circ$. The exterior angle at C is the angle between BC and the line AC extended beyond C. It lies on a straight line with $\angle C$, so it equals $180^\circ - \angle C$, which is $\angle A + \angle B$.

Fact 3 (isosceles triangles).
If two sides of a triangle are equal, then the angles opposite them are equal. Conversely, if two angles are equal, then the sides opposite them are equal.

If $AB = AC$, fold the triangle along the line bisecting $\angle A$. The equal sides AB and AC land on each other, so B lands on C and the triangle is symmetric; hence $\angle B = \angle C$. If the angle between the equal sides (the *apex* angle) is θ, each base angle is $\frac{1}{2}(180^\circ - \theta)$. Each angle of an equilateral triangle is 60°. (The converse is proved by folding along the perpendicular bisector of BC instead.)

Contest questions rarely say isosceles: equal sides come from one regular polygon, from two shapes that share an edge (as in the opening problem), or from lengths marked equal.

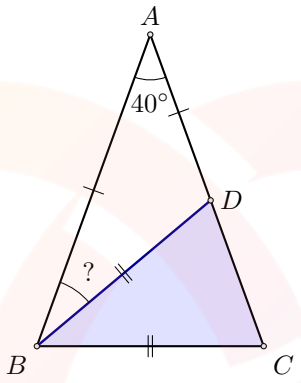


Example 1.
Lines ℓ and m are parallel. A point P between them is joined to A on ℓ and to B on m, as

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shown. The angle between AP and ℓ is $40°$, and the angle between BP and m is $65°$. Find $x = \angle APB$.

Solution.
No triangle contains x together with the given angles, and only Fact 1 links angles at different points on parallel lines. So we create a crossing at P: draw the line n through P parallel to ℓ and m.
The line n splits x into two parts. The upper part and the $40°$ angle are alternate angles between the parallel lines ℓ and n, so the upper part is $40°$. Similarly the lower part is alternate to the $65°$ angle, so it is $65°$. Hence $x = 40° + 65° = 105°$.
In general, when a path bends once between two parallel lines like this, the angle at the bend is the sum of the angles the path makes with the two lines.



Example 2.
Triangle ABC has $AB = AC$ and $\angle BAC = 40°$. The point D lies on AC with $BD = BC$. Find $\angle ABD$.

Solution.
Only one angle of triangle ABD is known, so we start in triangle ABC. It is isosceles with apex angle $40°$, so its base angles are $\angle ABC = \angle ACB = \frac{1}{2}(180° - 40°) = 70°$.
Triangle BCD has $BD = BC$, so its angles at D and C are equal. Since D lies on AC, $\angle BCD$ is the same angle as $\angle ACB$; this links the two triangles. So $\angle BDC = \angle BCD = 70°$, and $\angle DBC = 180° - 140° = 40°$. Therefore $\angle ABD = 70° - 40° = 30°$.
Check: $\angle BDC$ is the exterior angle of triangle ABD at D, so it should equal $40° + 30° = 70°$, and it does.

1.4 Polygons

Fact 4 (polygons).
The interior angles of an n-sided polygon add to $(n - 2) \times 180°$. The exterior angles of a convex polygon add to $360°$.

A polygon is *convex* if each interior angle is less than $180°$. For the first part, join one vertex to every other vertex. This cuts a convex n-sided polygon into $n - 2$ triangles (one for each side not touching that vertex), whose angles together make up the angles of the polygon. The *exterior angle* at a vertex is the angle between one side and the next side extended; it lies on a straight line with

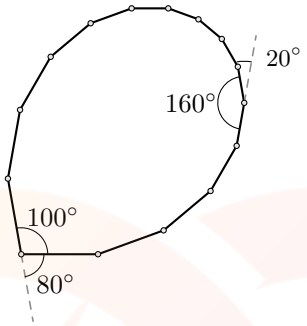
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the interior angle, so the two add to $180°$. For the second part, walk once round the boundary. At each corner you turn through the exterior angle, and you finish facing the way you started, having made one full turn.

A regular polygon has all its sides equal and all its angles equal, so

$$\text{exterior angle} = \frac{360°}{n}, \qquad \text{interior angle} = 180° - \frac{360°}{n}.$$

Know the interior angles $60°, 90°, 108°, 120°, 135°, 144°, 150°$ for $n = 3, 4, 5, 6, 8, 10, 12$. With regular polygons, work with the exterior angle first, since it is a single division.

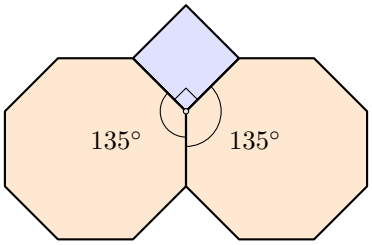


Example 3.
Every interior angle of a convex polygon is $160°$, except for one angle, which is $100°$. How many sides does the polygon have?

Solution.
Interior angles would give the equation $(n-1) \times 160° + 100° = (n-2) \times 180°$, but exterior angles are quicker. The exterior angle is $20°$ at each ordinary corner and $80°$ at the special one. The exterior angles add to $360°$, so the $20°$ angles add to $280°$, and there are $280 \div 20 = 14$ of them. The polygon has 15 sides.
Check: $14 \times 160° + 100° = 2340°$, and $(15-2) \times 180° = 2340°$.

1.5 Tilings

Regular polygons fit together round a point, with no gaps and no overlaps, exactly when their angles there add to $360°$. Copies of one regular polygon can do this only if its interior angle divides $360°$, as for triangles ($6 \times 60°$), squares ($4 \times 90°$) and hexagons ($3 \times 120°$). Three regular pentagons leave a gap of $36°$. A regular polygon with more than six sides has interior angle between $120°$ and $180°$, so two copies leave a gap and three overlap. Hence triangles, squares and hexagons are the only regular polygons that tile on their own.

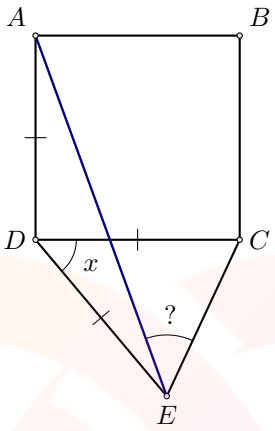


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Different regular polygons can share a point: a square and two regular octagons, as shown, since $90^\circ + 135^\circ + 135^\circ = 360^\circ$, or an equilateral triangle and two regular 12-gons, since $60^\circ + 150^\circ + 150^\circ = 360^\circ$.

1.6 When an angle cannot be found directly

If the information does not fix every angle in the diagram, call one unknown angle x, write the others in terms of x, and chase. Either x cancels, or the question needs more information.



Example 4.
$ABCD$ is a square, and E is a point on the other side of the line CD from A, such that $DE = DC$ and $\angle CDE$ is acute. Find $\angle AEC$.

Solution.
Nothing in the question fixes $\angle CDE$, so we call it x and expect it to cancel. The equal lengths are $DA = DC = DE$. Triangle DCE is isosceles with apex angle x, so $\angle DEC = \frac{1}{2}(180^\circ - x) = 90^\circ - \frac{1}{2}x$. Triangle DAE is isosceles with apex angle $\angle ADE = 90^\circ + x$, so $\angle DEA = \frac{1}{2}(90^\circ - x) = 45^\circ - \frac{1}{2}x$. From the diagram,

$$\angle AEC = \angle DEC - \angle DEA = (90^\circ - \tfrac{1}{2}x) - (45^\circ - \tfrac{1}{2}x) = 45^\circ.$$

In a multiple-choice question, choosing $x = 60^\circ$ (which makes DCE equilateral) gives a quick check, but only the argument with x shows that every allowed position of E gives 45°.

Common traps.

- Trusting the picture. Lines are parallel, or sides equal, only if the question says so or you can prove it.
- Mixing up interior and exterior angles, or treating the sum $(n-2) \times 180^\circ$ as a single angle.
- Forgetting the second case: an isosceles triangle with an angle of 50° could have it at the apex or at the base.

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1.7 How to spot this in a contest

1. If regular polygons appear, write down their interior angles first, and mark every side equal to the side they share.
2. Look for isosceles triangles before anything else.
3. If a path bends between parallel lines, draw an extra parallel line through the bend.
4. If an angle seems impossible to find, call one unknown angle x and expect it to cancel.
5. If the question asks for the number of sides of a polygon, use exterior angles.
6. Before answering, check that the angles of each triangle you used add to $180°$.

2 Practice problems

The problems are roughly in order of difficulty, from $\star$ to $\star\star\star$. Hints are in Section 4 and full solutions in Section 5.

Problem 1 $\star$ Each interior angle of a regular polygon is $140°$. How many sides does it have?

Problem 2 $\star$ $ABCD$ is a trapezium with AB parallel to DC. The sides AD and DC are equal, and $\angle ADC = 110°$. Find $\angle CAB$.

Problem 3 $\star\star$ In triangle ABC, the point D is the midpoint of BC, and $DA = DB$. Explain why $\angle BAC = 90°$.

Problem 4 $\star\star$ A square, a regular pentagon and a regular n-sided polygon, all with the same side length, fit together exactly round a point, with no gaps and no overlaps. Find n.

Problem 5 $\star\star\star$ Explain why a convex polygon can have at most three acute interior angles.

Problem 6 $\star\star\star$ A, B and C are three consecutive vertices of a regular polygon. An equilateral triangle ABP is drawn inside the polygon. Given that $\angle BPC = 40°$, how many sides does the polygon have?

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **UKMT JMC 2015 Question 6** $\star$
   Find the angles inside the triangle from the angles outside it.
2. **AMC 8 2017 Problem 6** $\star$
   Write the three angles as multiples of one unknown.
3. **UKMT JMC 2018 Question 3** $\star$
   What angle does a diagonal of a square make with its sides?
4. **UKMT Junior Kangaroo 2018 Question 7** $\star\star$
   Two isosceles triangles: call one base angle x.

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5. **AMC 8 2000 Problem 13** ⋆⋆
   Base angles of the isosceles triangle first, then the bisector.

6. **UKMT Junior Kangaroo 2017 Question 13** ⋆⋆
   Co-interior angles along each of the two sides that are not parallel.

7. **UKMT JMC 2017 Question 11** ⋆⋆
   Add up all the angles at all seven vertices together.

8. **UKMT JMC 2019 Question 21** ⋆⋆
   Sides of different polygons are equal: look for an isosceles triangle.

9. **AMC 8 2016 Problem 23** ⋆⋆⋆
   Join E to both centres; all the radii are equal.

10. **UKMT JMC 2015 Question 25** ⋆⋆⋆
    Exterior angles of two triangles, then the isosceles triangle.

11. **UKMT JMC 2018 Question 25** ⋆⋆⋆
    Find the angle of the polygon from the angle sum of the quadrilateral.

12. **UKMT JMC 2019 Question 24** ⋆⋆⋆
    Look for two congruent triangles and equal exterior angles.

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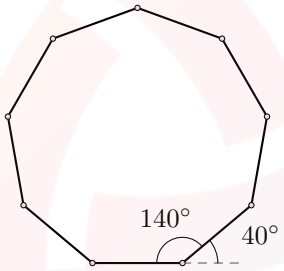
4 Hints

1. Find the exterior angle first.
2. Which triangle has two equal sides? After that, use the parallel lines.
3. There are two isosceles triangles. Call their base angles x and y.
4. The three angles at the point add to $360°$. Find the third interior angle, then its exterior angle.
5. Look at the exterior angles. What can you say about the exterior angle at an acute corner?
6. Which two sides of triangle PBC are equal?

5 Solutions

Problem 1

Each interior angle of a regular polygon is $140°$. How many sides does it have?



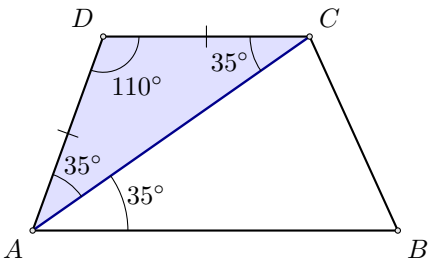
At each vertex the interior angle and the exterior angle lie on a straight line, so each exterior angle is $180° - 140° = 40°$. The exterior angles of a polygon add to $360°$, and in a regular polygon they are all equal, so there are $360 \div 40 = 9$ of them. The polygon has 9 sides.

As a check, the interior angles add to $9 \times 140° = 1260°$, and $(9 - 2) \times 180° = 1260°$.

Answer: 9

Problem 2

$ABCD$ is a trapezium with AB parallel to DC. The sides AD and DC are equal, and $\angle ADC = 110°$. Find $\angle CAB$.



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Since $DA = DC$, triangle ADC is isosceles, and its base angles are $\angle DAC = \angle DCA = \frac{1}{2}(180^\circ - 110^\circ) = 35^\circ$.

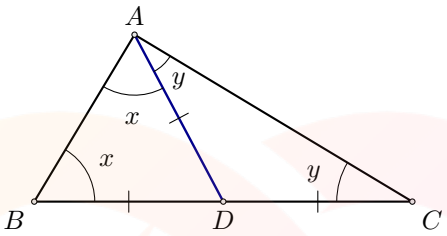
The diagonal AC crosses the parallel lines AB and DC, and $\angle CAB$ and $\angle DCA$ are alternate angles (the Z shape). Hence $\angle CAB = \angle DCA = 35^\circ$.

Note that AC bisects $\angle DAB$, since both halves are 35°. The same argument shows this in every trapezium with $AD = DC$.

Answer: 35°

Problem 3

In triangle ABC, the point D is the midpoint of BC, and $DA = DB$. Explain why $\angle BAC = 90^\circ$.



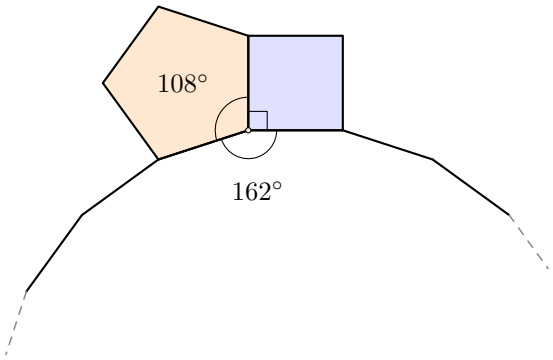
Since D is the midpoint of BC, we have $DB = DC$, and so $DA = DB = DC$. Triangle ABD is isosceles with $DA = DB$, so its base angles are equal; call them x, so that $\angle DBA = \angle BAD = x$. Triangle ADC is isosceles with $DA = DC$; call its base angles y, so that $\angle DCA = \angle CAD = y$.

The angles of triangle ABC are $\angle B = x$, $\angle C = y$ and $\angle A = \angle BAD + \angle CAD = x + y$. They add to 180°, so $2x + 2y = 180^\circ$, which gives $x + y = 90^\circ$. Hence $\angle BAC = x + y = 90^\circ$.

This fact is worth remembering: if the midpoint of one side of a triangle is the same distance from all three vertices, the angle opposite that side is a right angle. At Level 2 it reappears as the theorem that the angle in a semicircle is a right angle.

Problem 4

A square, a regular pentagon and a regular n-sided polygon, all with the same side length, fit together exactly round a point, with no gaps and no overlaps. Find n.



The angles at the point add to 360°. The square contributes 90°. The interior angles of a pentagon add to $3 \times 180^\circ = 540^\circ$, so each angle of the regular pentagon is $540^\circ \div 5 = 108^\circ$. Hence the third polygon has interior angle $360^\circ - 90^\circ - 108^\circ = 162^\circ$.

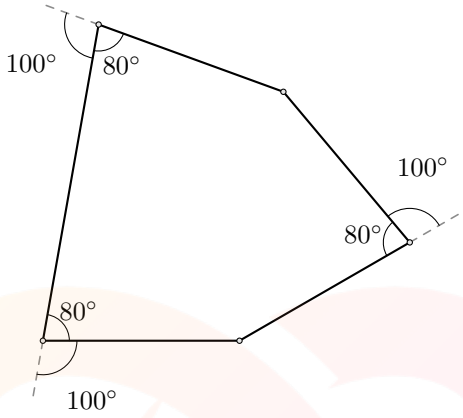
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Its exterior angle is $180° - 162° = 18°$, and the exterior angles add to $360°$, so it has $360 \div 18 = 20$ sides.

Answer: 20

Problem 5

Explain why a convex polygon can have at most three acute interior angles.



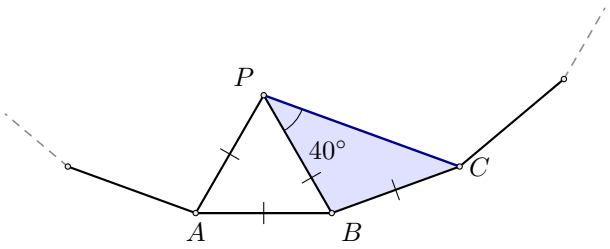
At each vertex, the interior and exterior angles add to $180°$. So if the interior angle is acute (less than $90°$), the exterior angle is more than $90°$. In a convex polygon every exterior angle is positive, and together they add to $360°$.

Suppose the polygon had four or more acute angles. The exterior angles at four of them would add to more than $4 \times 90° = 360°$, and all the other exterior angles are positive, so the total of the exterior angles would be more than $360°$. This contradicts Fact 4. Hence there are at most three acute angles.

Three is possible: an equilateral triangle has three acute angles, and so does the pentagon in the diagram, whose angles are $80°$, $150°$, $80°$, $150°$ and $80°$. (They add to $540° = 3 \times 180°$, as they should.)

Problem 6

A, B and C are three consecutive vertices of a regular polygon. An equilateral triangle ABP is drawn inside the polygon. Given that $\angle BPC = 40°$, how many sides does the polygon have?



The sides BP and BA belong to the equilateral triangle, and BA and BC are sides of the regular polygon, so $BP = BA = BC$. Triangle PBC is therefore isosceles with $BP = BC$, and its base angles are equal: $\angle BCP = \angle BPC = 40°$. Its apex angle is $\angle PBC = 180° - 40° - 40° = 100°$.

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The interior angle of the polygon at B is made up of $\angle ABP = 60^\circ$, an angle of the equilateral triangle, and $\angle PBC = 100^\circ$, so it is 160°. The exterior angle is 20°, so the polygon has $360 \div 20 = 18$ sides.

As a check, in a regular 18-gon the interior angle is $180^\circ - 20^\circ = 160^\circ$, so $\angle PBC = 160^\circ - 60^\circ = 100^\circ$, and the base angles of triangle PBC are $\frac{1}{2}(180^\circ - 100^\circ) = 40^\circ$.

Answer: 18